

E payment system

Thursday, September 25, 2025

8:43 AM

Electronic Payment System (EPS)

Definition: EPS is a way to pay for goods and services **electronically** instead of using cash or physical cheques. Simply put, it means **online or digital transactions**.

Examples of EPS: Online credit card payments, digital wallets, mobile payments, e-checks, virtual currency, etc.

Why it's needed:

- Online shopping between individuals and businesses
- Mobile platforms make digital payments faster, safer, and convenient

Types of E-Payment Systems

1. Debit Card

- Linked to your bank account; money is deducted immediately.
- Can be used online or at ATMs.
- Requires a **PIN**.
- Example: Paying for groceries online using your bank debit card.

2. Credit Card

- Borrow money from the bank to pay now, repay later.
- Interest applies if not paid on time.
- Example: Using Visa or Mastercard to buy a laptop online and paying next month.

3. Smart Card

- Card with a chip storing your account info.
- Can be used for payments, authentication, or access control.
- Example: A bank-issued card used for secure office entry and payments.

4. E-banking / Mobile Banking

- Banking services online via computers or smartphones.
- Includes paying bills, transferring money, loans, or topping up wallets.
- Example: Paying your electricity bill using eSewa or Khalti.

5. E-cash

- Prepaid card issued by a bank.
- Money must be deposited before use; can also withdraw cash.
- Example: Prepaid cards used in cafes or online games.

6. E-cheque

- Digital version of a cheque with **digital signatures**.
- No authenticity issues like paper cheques.
- Example: Paying a supplier online using an e-cheque.

7. E-wallet / Digital Wallet

- Online/mobile account to store money securely.
- Can pay for shopping, bills, top-up mobile, etc.
- Example: **Khalti, eSewa, PayPal**

Online Credit Card Transactions

Definition: Buying goods or services online using a credit card. Also called **Cardholder Not Present (CNP) transactions** because the merchant never sees the actual card and no signature is taken.

Parties Involved (5 main ones)

1. **Consumer** – The buyer.
2. **Merchant** – The online store.
3. **Clearinghouse** – Middleman that processes the transaction.
4. **Merchant Bank** – Receives payment for the merchant.
5. **Card-Issuing Bank** – The bank that issued the consumer's card.

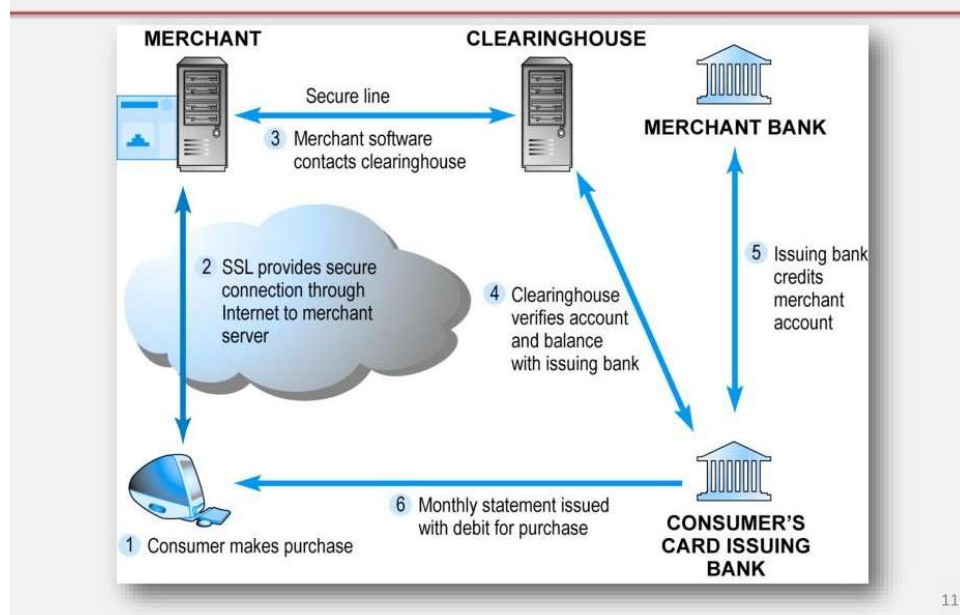
How It Works (Step-by-Step)

1. **Purchase** – Consumer adds items to the online shopping cart and proceeds to pay.
2. **Secure Connection** – SSL/TLS creates a safe “tunnel” to protect card info over the Internet.
3. **Merchant Contacts Clearinghouse** – Merchant software sends card details to the clearinghouse.
4. **Verification** – Clearinghouse checks with the issuing bank to confirm account and balance.
5. **Payment to Merchant** – Issuing bank approves and credits the merchant's account.
6. **Consumer Billing** – The purchase is added to the consumer's monthly credit card statement.

Limitations

- **Security Risks** – Card details can be stolen if the connection or merchant site isn't secure.
- **Social Equity Issues** – Not everyone has access to credit cards, limiting who can shop online.

How an Online Credit Transaction Works



Online Stored Value Payment System (SVPS)

Definition:

A system where users deposit money into an account (or card) and spend it whenever needed. Mainly used for **small, low-value transactions**.

- **Advantages:** Low transaction cost, no need for bank verification during payment.
- **Example:** PayPal, Google Pay balance.

Types of Stored Value Cards

1. Closed System Prepaid Cards

- Can only be used at **one specific merchant**.
- Replace traditional gift certificates.
- Often have expiration dates or service fees.

- **Example:** A gift card issued by Bhat-Bhateni that can only be used in their stores.

2. Open System Prepaid Cards

- Can be used anywhere that accepts the card's network (like Visa or Mastercard).
- No credit is given; users can only spend what they load onto the card.
- Uses **magnetic stripe or digital storage** for the prepaid balance.
- **Example:** A Visa prepaid card or Mastercard gift card.

Social/Mobile Peer-to-Peer (P2P) Payment System

Definition:

P2P payment allows individuals to send money directly to friends, family, or anyone else online using **bank accounts, credit/debit cards, or mobile apps**.

- No need for cash or physical wallets.
- **Examples:** PayPal, Google Pay, eSewa, Khalti.

Key Point: Makes everyday payments faster and easier using just a smartphone.

Electronic Billing Presentment & Payment (EBPP) Systems

Definition:

EBPP lets consumers **view and pay bills online** instead of using paper bills.

- Bills are delivered electronically.
- Payments are made via **bank transfers or credit/debit cards**.
- Example: Paying electricity bills in Nepal using eSewa or a banking app, with an **electronic receipt**.

EBPP Business Models:

1. **Online Bank Direct:** Bills sent and paid directly via bank websites.
 2. **Mobile:** Bills delivered and paid via mobile apps.
 3. **Consolidators:** Third-party platforms collect multiple bills for easier payment.
- **Most common:** Online bank direct model.

Auctioning in E-commerce

Definition:

Auctioning is selling goods, assets, or resources online through **competitive bidding**. It can happen:

- **B2B** (business to business)
- **B2C** (business to consumer)
- **C2C** (consumer to consumer)

Purpose:

- Helps discover the real market price of an item.
- Sellers can get **competitive value** for assets.
- Bidders can join **virtually from anywhere**.

Types of Auctions

1. English Auction (Most common)

- Price starts low and **increases** with each bid.
- **Single item sold**.
- **Highest bidder wins**.
- Used when seller **doesn't know the market price**.

2. Dutch Auction

- Price starts high and **decreases** until a buyer accepts.
- Can sell **single or multiple items**.
- **First bidder wins**.
- Used when seller **has an expected price**.

3. Double Auction

- Buyers place bids and sellers place offers simultaneously.
- Can sell **single or multiple items**.
- **First matching bid wins**.
- Used in markets with **multiple buyers and sellers**, e.g., stock markets.

4. Vickery Auction (Sealed-bid)

- Bidders submit bids **without seeing others' bids**.
- **Highest bidder wins**, but pays the **second-highest bid**.

SET Protocol (Secure Electronic Transaction)

Definition:

SET is a **communication protocol** that makes **electronic payments (like credit cards) secure**. It is **not a payment system itself**, but ensures transactions are safe on open networks using **encryption and cryptography**.

Features of SET Protocol

1. Merchant Authentication

- Confirms the merchant is legitimate.
- Uses **X.509V3 certificates** to verify merchant's relationship with financial institutions.

2. Cardholder Authentication

- Ensures the credit card is being used by the **authorized owner**.
- Verified through **X.509V3 certificates**.

3. Message Confidentiality

- Prevents unauthorized people from reading transaction messages.
- Achieved using **encryption**, usually **DES**.

4. Message Integrity

- Prevents messages from being **altered or tampered with**.
- Achieved using **RSA digital signatures** or **HMAC with SHA-1**.

SET Participants

1. **Cardholder** – Person who owns the credit card (Visa, MasterCard) issued by a bank.
2. **Merchant** – Seller who accepts credit card payments. Must have a relationship with an **Acquirer**.
3. **Issuer** – Bank that issues the card to the cardholder.
4. **Acquirer** – Bank that holds the merchant's account and processes credit card payments.
5. **Certificate Authority (CA)** – Issues **X.509V3 certificates** to cardholders, merchants, and payment gateways for authentication.
6. **Payment Gateway** – Connects merchant website to the banking/payment system. Acts as a secure interface.

How SET Protocol Works (Step-by-Step)

1. Customer browses a merchant's website and chooses products.
2. Customer sends **order + payment info** (purchase order + card info).
3. Merchant forwards **card info** to their bank (Acquirer).
4. Acquirer contacts the **Issuer** for payment authorization.
5. Issuer verifies and sends **authorization** back to Acquirer.
6. Acquirer sends **authorization** to the merchant.
7. Merchant completes the order and sends confirmation to the customer.
8. Merchant captures the payment from their bank.
9. Issuer includes the charge in the **customer's credit card bill**.

SET Protocol Phases:

1. Cardholder Registration

- Cardholder (C) sends credit card info to **Certification Authority (CA)**.
- CA verifies the card and issues a **digital signature certificate**.
- Cardholder stores it for future secure transactions.

(Protects sensitive credit card info.)

2. Merchant Registration

- Merchant (M) registers **public keys for signature and encryption** with CA.
- Process is shorter since there's no confidential card info.

3. Purchase Request

- Cardholder chooses products and sends **order + payment info** to Merchant.
- Merchant processes order and forwards payment info to **Payment Gateway (PG)**.

4. Payment Authorization

- PG checks payment details with **Issuer and banks**.
- If approved, PG sends **authorization** to Merchant.
- Merchant confirms order to Cardholder and delivers goods/services.

5. Payment Capture

- Merchant sends **payment requests + capture tokens** to PG.
- PG verifies and confirms.
- Actual money transfer from Cardholder to Merchant occurs outside the protocol.

Dual Signature

- Introduced with **SET** for privacy and security.
- Purpose: Keep **Order Information (OI)** and **Payment Information (PI)** separate.
 - **OI → Merchant** (what you're buying, quantity, etc.)
 - **PI → Bank** (credit card or payment details)
- Benefit: Merchant **doesn't see your card number**, and bank **doesn't see what you're buying**.
- Ensures privacy and secure handling of sensitive data.

Example:

- You buy Coke from Bhatbhateni online.
 - Merchant sees: "2 bottles of Coke, delivery address."
 - Bank sees: "Payment of \$4 from your credit card."

Virtual Currency

- Digital money **only exists electronically**.
- Stored and transacted via software, apps, or online platforms.
- Can be issued by private companies or developers.
- Mostly **unregulated**, unlike government-backed currencies.
- Advantages: Fast transactions, convenient.
- Disadvantages: Can be hacked, limited legal protection.

Example:

- Bitcoin, Ethereum, or store-specific tokens used for online purchases.

Status of E-Payment Systems in Nepal

- With internet growth, online payments are becoming common in Nepal.
- Popular services include **eSewa, Khalti, and mobile banking apps**.
- People use these for:
 - Sending/receiving money
 - Paying utility bills
 - Buying tickets or mobile recharge
- **Digital wallet users:** ~4 million in Nepal.
- Trend: From ATM cards → e-banking → mobile wallets.
- **Digital transactions growth:**
 - 2018: NPR 712 billion
 - 2019: NPR 1,559 billion
- **Advantages of e-payment systems:**
 - Lower transaction costs
 - Secure payments

- Saves time and resources
- Faster payments
- Transparency and visibility throughout transactions

Properties of E-Cash

1. **Monetary Value:** Must be backed by bank-certified cash, credit cards, or checks.
2. **Interoperability:** Should be exchangeable for other e-cash, goods, services, or paper cash.
3. **Retrievability:** Can be stored and retrieved for future use.
4. **Security:** Must be hard to copy or tamper with.